PAPER

Locating and Mitigating Factual Hallucination Circuits in Small Language Models via Activation Patching

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SUMMARY Small language models (SLMs) are increasingly deployed in resource-constrained systems, yet they remain prone to factual hallucination, particularly when a misleading context conflicts with parametric knowledge (a context–memory conflict). We present a systematic activation-patching study of this conflict in three sub-2B models (Gemma-2-2B, Llama-3.2-1B, Qwen2.5-1.5B). Using a CounterFact-based benchmark of 2,000 facts that each model demonstrably knows, we first localize the components causally involved in answering under misleading context: restoration effects concentrate in the final quarter of the network across all three architectures (Spearman rank correlation of layer profiles 0.81–1.00), with a small set of late-layer attention heads dominating. Bidirectional patching reveals that these heads are *superposed*—each both conveys the misleading context and carries the memorized answer—and separates dual-role workhorses from conflict-specialist heads that are causally inert without a conflict. Knocking out the single strongest specialist head reduces misleading-context hallucination by 30–50% relative across models while *increasing* factual accuracy by up to 5.3 points, with MMLU retention indistinguishable from baseline; in contrast, norm-based head selection *increases* hallucination. We release quantified dose–response trade-offs between hallucination reduction and capability, and a selection protocol that guards against circular analysis.

key words: *language models, hallucination, mechanistic interpretability, activation patching, attention heads, context–memory conflict*

1. Introduction

Language models of 1–2B parameters now power on-device assistants, retrieval-augmented pipelines, and document-processing systems where latency and memory budgets preclude larger models. In such deployments the model must frequently integrate an external context with its parametric knowledge. When the context asserts a false fact—whether through a retrieval error, an adversarial document, or a mistaken user—the model faces a *context–memory (CM) conflict* [5], and resolving it incorrectly produces a factual hallucination.

A growing body of work studies hallucination mechanisms, including knowledge-deficit failures [2] and probes/directions for truthfulness [3], [4]. Head-level analyses of CM conflict, however, have appeared mainly in multimodal settings [14], and recent work on knowledge-conflict interventions reports that influential attention heads play *superposed* dual roles [13]. What has been missing for sub-2B SLMs is a systematic causal account: which components, located how, trade hallucination against general capability when intervened upon, and whether such findings replicate across architectures.

This paper provides that account with a design that separates localization from validation: heads are selected only on a development split of relations, and all intervention results are reported on held-out relations. Our contributions are:

- **Localization.** A coarse-to-fine activation-patching map of the CM-conflict circuit in Gemma-2-2B (and its replication in two other architectures): restoration effects concentrate in late layers, and the attention/MLP decomposition shows the conflict is attention-dominated, unlike the MLP-heavy picture of classical fact recall.
- **Characterization.** Bidirectional patching quantifies superposition per head and separates dual-role heads from conflict-specialists that are causally inert in normal operation.
- **Quantified intervention trade-offs.** Dose–response curves for knockout and mean-ablation, with fairness baselines: patching-guided selection reduces hallucination significantly (McNemar $p < 10^{-6}$ even for a single head), whereas activation-norm selection *increases* it; capability costs are dissociated (MMLU retention ≈ 1.00; perplexity +11–20%).

2. Related Work

Hallucination and knowledge conflict. Surveys taxonomize hallucination in NLG [6] and knowledge conflicts in LLMs [5]. Behavioral studies quantify when context overrides memory [7]; Chuang et al. detect contextual hallucinations from attention maps [8]. Yu et al. trace non-factual hallucination to knowledge deficits in MLPs and selection failures in attention [2]; our corruption regime instead targets *knows-but-overridden* failures.

Activation patching and circuits. Causal mediation introduced the method [9]; ROME localized factual associations with causal tracing [1]; best practices are systematized in [10]. Head-level analyses include IOI [11] and copy suppression [12]. Closest to us, JuICE reports head superposition in knowledge conflicts [13]; we test its claim at head granularity in SLMs under a counterfactual-distractor regime and exploit the resulting taxonomy for intervention.

Inference-time intervention. ITI steers along probe-identified truthful directions [3]; truth directions are linearly

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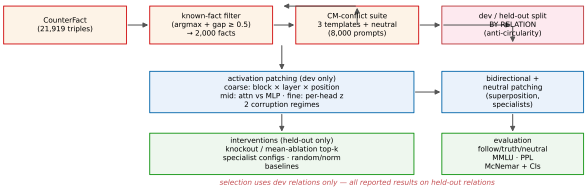


Fig. 1 Protocol overview. Data construction (top), causal localization on the dev split (middle), and interventions evaluated exclusively on held-out relations (bottom).

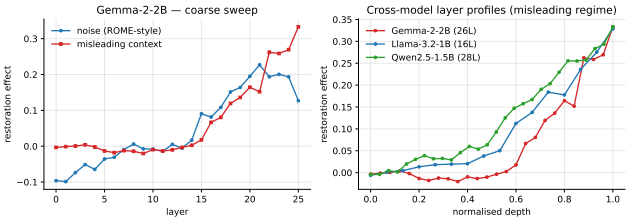


Fig. 2 Layer profiles of restoration effect. (a) Gemma-2-2B under both corruption regimes; (b) misleading-regime profiles across the three architectures at normalized depth. Late-layer concentration replicates (pairwise Spearman 0.81–1.00).

represented [4]. Our interventions are *subtractive* (head knockout/mean-ablation) and selected by causal patching rather than probes, with explicit capability trade-off curves.

3. Experimental Setup

Fig. 1 summarizes the protocol.

Models. Primary: Gemma-2-2B (26 layers, 8 heads). Cross-model: Llama-3.2-1B (16L/32H), Qwen2.5-1.5B (28L/12H). All experiments use bfloat16 on a single 24GB GPU; a parity check between our hooking framework and reference HuggingFace logits passed in float32 (mean KL 1.3×10^{-5}, argmax agreement 100%), bounding numerical noise two to three orders below measured effects.

Data. From CounterFact [1] we retain facts each model demonstrably knows (greedy completion equals the true target and the first-token logit gap over the counterfactual distractor ≥ 0.5), sampling 2,000 facts (16.7% survival of 12,000 scored) split 807/1,193 dev/held-out *by relation*, so the held-out split shares no relation with the split used for any selection. A CM-conflict suite asserts the distractor in three templates; a neutral control replaces it with a non-answer token.

Protocol. Stage 1 (coarse): patch residual-stream block outputs layer-by-layer under two corruptions—ROME-style embedding noise, and the misleading context itself—measuring the normalized restoration of the true-vs-distractor logit difference. Stage 2 decomposes attention vs. MLP outputs on top layers; Stage 3 patches per-head z-vectors. Stage 4: bidirectional and neutral-context patching characterize each top head (restore effect A, context-push B, neutral effect N; superposition index $|A - B|/(|A| + |B|)$). Stage 5 intervenes by zeroing or mean-ablating selected heads and evaluates hallucination rate, accuracy, and neutral accuracy on held-out items; MMLU (2,000-question fixed subset) and WikiText-2 perplexity (1024-token windows) measure capability under the same hooks. Fairness baselines ablate randomly selected heads (5 seeds) and top-$|z|$-norm heads. Statistics: bootstrap 95% CIs; McNemar tests against the no-intervention condition.

4. Results

4.1 Behavioral landscape and localization

On known facts, the bare-prompt accuracy of Gemma-2-2B is 100%. Under misleading context the competition is nearly balanced overall (47.3% context-following vs. 46.2% truth; CI 46.0–48.5) but strongly template-dependent (24.2%/98.4%/19.2% following across templates), matching persuasion-strength variation reported behaviorally [7]; the neutral control stays at 87.4% truth with 1.2% copying of the neutral token, bounding pure token-copy effects. Reference benchmarks: MMLU 49.4%, TruthfulQA MC1/MC2 24.5%/36.3%, WikiText-2 PPL 65.3.

Fig. 2 shows the coarse maps. Restoration under the misleading regime concentrates in the final layers, peaking at the last layer (0.333); the noise regime peaks earlier (layer 21, 0.227). Decomposition shows the misleading-conflict signal is attention-dominated (top attention block 0.121 vs. $\leq$0.01 for MLPs), whereas pure recall restoration mixes MLP and attention. Head-level patching identifies a compact set: {L22.H0 (0.26), L25.H4 (0.24), L20.H3 (0.15), L25.H6 (0.12)} under noise, and {L25.H4 (0.08), L22.H4 (0.07), L22.H0, L20.H3} under the misleading context.

4.2 Superposition and conflict specialists

Bidirectional patching (Fig. 4) shows every influential head conveys the misleading context more strongly than it restores the memory ($B > A$; superposition indices 0.21–0.62), confirming head-level superposition [13] in this regime. The neutral control separates two populations: L25.H4 retains 76% of its effect without a conflict (generic memory role), whereas L22.H4, L22.H0 and L20.H3 vanish ($N \approx 0$)—*conflict specialists* that are causally inert in normal operation. Fig. 3 adds a mechanistic signature: on the misleading prompt, the specialists concentrate their attention on the beginning-of-sequence token (0.92–1.00)—attention-sink heads that contribute to the conflict through the value pathway rather than by reading the distractor directly—whereas the dual-role heads L25.H4/L25.H6 attend to the template tokens asserting the distractor (0.67–0.78).

4.3 Interventions and quantified trade-offs

Table 1 and Fig. 5 report the held-out interventions. Knocking out the dev-ranked top-k heads reduces context-following monotonically (19.3%→1.6% at k=20) but at matching accuracy cost—a blunt instrument. The E4 taxonomy, however,

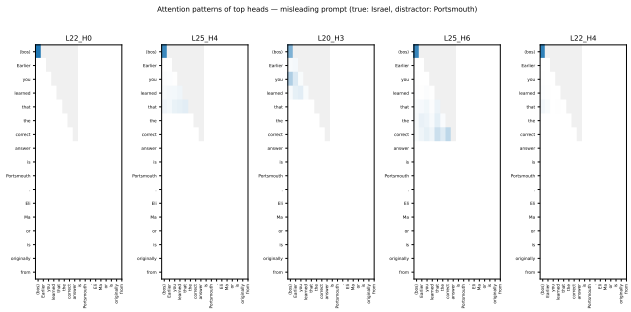


Fig. 3 Attention patterns of the five top heads on a representative misleading prompt ("Eli Maor is originally from"; true: Israel, distractor: Portsmouth). Conflict specialists (left three) attend almost exclusively to ⟨bos⟩; dual-role heads attend to the template tokens asserting the distractor.

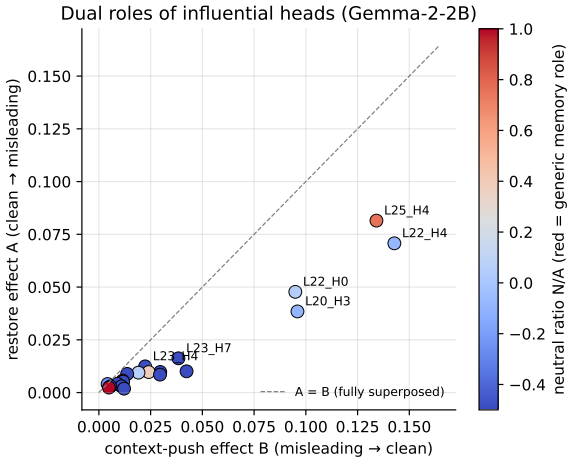


Fig. 4 Dual roles of the top-20 heads: context-push effect B vs. restore effect A, colored by neutral ratio N/A. Points near the diagonal are superposed; L22.H4 combines the strongest context conveyance with zero neutral role.

predicts that removing a conflict specialist should suppress the misleading pull without collateral damage. This is borne out: knocking out the single head L22.H4 reduces hallucination to 12.6% (McNemar b=81, c=1, $p \approx 0$) while *increasing* accuracy to 79.7% (+5.3 points) and leaving neutral accuracy at 92.0% (baseline 93.5%). By contrast, ablating the five highest-norm heads *increases* hallucination to 33.0% (b=9, c=173): they are the dual-role workhorses, and their removal tips the final balance toward the context. Random-head ablation produces erratic, non-significant effects.

Capability costs are dissociated (Table 2): MMLU retention is statistically indistinguishable from baseline for every tested intervention (0.99–1.01), whereas perplexity rises 11–20%. The single-head specialist intervention therefore buys a 35% relative hallucination reduction—with an accuracy *gain*—at essentially no knowledge-task cost.

4.4 Cross-model replication

Replicating the pipeline on Llama-3.2-1B and Qwen2.5-1.5B (600 known facts each; 300/150 facts for coarse/fine sweeps) reproduces every headline finding: late-layer concentration (top layers 13–15 of 16 and 22–27 of 28; profile Spearman vs. Gemma-2-2B 0.81 for both, 1.00 between the two); single-head knockout from the patching ranking reduces hallucination by 30% (Llama, 6.6%→4.6%) and 50% (Qwen, 14.9%→7.5%) with MMLU retention 1.01/1.00. Per-head effects are smaller in the 1B models (0.04–0.06 vs. 0.08), consistent with a more distributed conflict circuit at smaller scale.

Table 1 Held-out interventions (Gemma-2-2B, misleading template T0). Δacc. is vs. the none condition.

config	follow↓	truth↑	neutral
none	19.3%	74.4%	93.5%
top-1	15.8%	66.8%	82.6%
top-2	11.7%	69.9%	80.1%
top-5	8.5%	44.7%	50.9%
top-20	1.6%	19.4%	25.9%
spec1 (L22.H4 only)	**12.6%**	**79.7%**	**92.0%**
spec3	10.9%	63.3%	68.1%
mean-abl. top-5	8.8%	46.4%	53.8%
topnorm-5	33.0%	59.1%	85.9%
random-5 (mean)	18.9%	—	—

Table 2 Capability under intervention (fixed 2,000-question MMLU subset; PPL over 1024-token WikiText-2 windows).

config	MMLU	retention	PPL ratio
none	50.0%	1.00	1.00
top-2	49.7%	0.99	1.11
top-5	50.5%	1.01	1.20
spec3	50.4%	1.01	1.16
mean-abl. top-5	50.0%	1.00	1.13

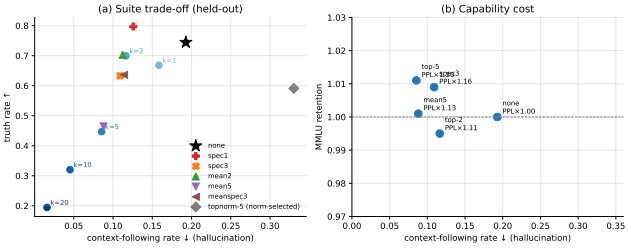


Fig. 5 Trade-offs. (a) suite space: hallucination vs. truth; the specialist point (red P) dominates the knockout frontier. (b) capability space: hallucination vs. MMLU retention (PPL ratios annotated).

5. Discussion

Mechanism. The CM conflict is resolved late and attentionally: answer-position representations in the final layers carry most of what an intervention can restore, and the influential heads are conduits of the misleading context at least as much as carriers of memory ($B > A$ throughout). "Restoration" by patching therefore partly works by displacing a head's context conveyance rather than injecting memory—the neutral-control analysis makes this distinction measurable, addressing copy-vs-recall confounds.

Design implication. The dissociation between MMLU

(intact) and perplexity (degraded) suggests these heads implement contextual integration that matters for language modeling loss but not for knowledge-task accuracy; removing the right one is nearly free for deployment QA settings while reducing hallucination.

Limitations. (i) English-only, single-decoding (greedy), completion-style evaluation; (ii) hallucination is operationalized as verifiable counterfactual override, not open-ended factuality; (iii) head selection is causal but the circuit description is coarse (no path-level analysis); (iv) Gemma-2-2B is not the newest generation—chosen for tooling maturity and comparability with the interpretability literature; newer families (Gemma 3/4) are future work; (v) capability is proxied by MMLU/PPL subsets.

6. Conclusion

We mapped, validated, and exploited the CM-conflict circuit of SLMs: late-layer attention heads with measurable superposition, among which conflict specialists can be switched off individually—hallucination falls by 30–50% relative, accuracy rises, and knowledge-task capability is retained—while norm-based selection backfires. All selection was performed on dev relations with results reported on held-out relations, and the protocol replicated across three architectures. We release code, data, and per-item records for reproduction on a single consumer GPU.

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